

Pranav Nuti

The University of Chicago Booth School of Business
☎ +1 650 283 9394 • ✉ pranav.nuti@chicagobooth.edu
🌐 pranavnuti.com

Employment

The University of Chicago Booth School of Business 2024–
Principal Researcher
Postdoctoral-level position, mentored by Rad Niazadeh

Education

Stanford University 2018–2024
PhD in Mathematics
Advised by Jan Vondrák

Indian Institute of Science (IISc), Bangalore 2014–2018
Bachelor of Science (Research), Gold Medalist in Mathematics

Research

Journal Publications and Revisions.....

Prophet Inequalities with Cancellation Costs <i>with Farbod Ekbatani, Rad Niazadeh, and Jan Vondrák</i>	Operations Research, Minor Revision
Toward an Optimal Contention Resolution Scheme for Matchings <i>with Jan Vondrák</i>	Mathematical Programming
Online Matching and Contention Resolution for Edge Arrivals with Vanishing Probabilities <i>with Will Ma and Calum MacRury</i>	Operations Research, Minor Revision
The Secretary Problem with Distributions	Mathematics of Operations Research, Major Revision

Peer-Reviewed Conference Publications.....

Stationary Online Contention Resolution Schemes <i>with Mohammad Reza Aminian and Rad Niazadeh</i>	EC 2026
Optimal Prophet Inequalities with Buyback <i>with Farbod Ekbatani, Rad Niazadeh, and Jan Vondrák</i>	EC 2026
Static Pricing for Single Sample Multi-unit Prophet Inequalities <i>with Peter Westbrook</i>	SOSA 2026
Online Matching and Contention Resolution for Edge Arrivals with Vanishing Probabilities <i>with Will Ma and Calum MacRury</i>	EC 2024
Prophet Inequalities with Cancellation Costs <i>with Farbod Ekbatani, Rad Niazadeh, and Jan Vondrák</i>	STOC 2024
Secretary Problems: The Power of a Single Sample <i>with Jan Vondrák</i>	SODA 2023
A Tight Competitive Ratio for Online Submodular Welfare Maximization <i>with Amit Ganz and Roy Schwartz</i>	ESA 2023

Toward an Optimal Contention Resolution Scheme for Matchings <i>with Jan Vondrák</i>	IPCO 2023
--	-----------

The Secretary Problem with Distributions	IPCO 2022
---	-----------

Working Papers.....

Single-Sample Prophet Inequalities for Knapsack Problems <i>with Rebecca Reiffenhäuser and Alice Sayutina</i>	(submitted)
---	-------------

Optimal Bayesian Online Allocation of Reusable Resources <i>with Mohammad Reza Aminian and Rad Niazadeh</i>	(submitted)
---	-------------

Workshop Papers.....

Solving Open Problems in Operations Research Using AI <i>with Eric Fithian and Rad Niazadeh</i> Draft paper. Based on Open Problems in Operations Research. Also presented as a poster at Can AI do Theory? at STOC 2026.	AI-Driven Research in EconCS at EC 2026
--	---

Bounded Best Buyers: Prophet Inequalities Revisited <i>with Farbod Ekbatani, Rad Niazadeh, and Jan Vondrák</i>	Easy Peasy at EC 2026
--	-----------------------

Magicians Don't Move: An Easy Peasy OCRS <i>with Mohammad Reza Aminian and Rad Niazadeh</i>	Easy Peasy at EC 2026
---	-----------------------

A Match Made by Entropy <i>with Mohammad Reza Aminian and Rad Niazadeh</i>	Easy Peasy at EC 2026
--	-----------------------

Selected Talks.....

How to Write and Review Papers <i>EC 2026 Mentoring Workshop</i>	2026
--	------

Prophet Inequalities with Cancellation Costs <i>Chicago Junior Theorists Workshop</i>	2025
---	------

What is a Good Model for Online Decision Making? <i>SURIM Talk, Stanford University</i>	2023
---	------

Online Selection Problems and How Ramsey's Theorem Solves a Card Game <i>DACO Seminar, ETH Zürich</i>	2023
---	------

Between the Secretary Problem and the Prophet Inequality <i>Data-Driven Decision Processes Talk, Simons Institute, UC Berkeley</i>	2022
--	------

Teaching.....

Papers.....

Engaging Activities for Teaching Linear Algebra <i>with Shintaro Fushida-Hardy and Megan Selbach-Allen</i>	PRIMUS 2024
--	-------------

Awards.....

Pólya Teaching Fellow Award <i>Awarded for outstanding teaching during the PhD program</i>	2024
--	------

Robert Osserman Teaching Award <i>Awarded for outstanding contributions as a teaching assistant at Stanford University</i>	2022
--	------

Hoefer Mentoring Award <i>Awarded for supporting undergraduate writing at Stanford University</i>	2022
---	------

Courses Taught as Instructor.....

Linear Algebra, Stanford Summer Engineering Academy 2021, 2022

Equity and Inclusion Initiative, Stanford Engineering

This is a summer program that seeks to illuminate the brilliance of students who have been systemically marginalized in engineering. Along with teaching the course, in collaboration with my co-instructors, I also designed the curriculum. 38 classroom hours. Syllabus, lesson plans, handouts and descriptions of all activities are available.

Teaching Outside the University.....

Stanford Jail and Prison Education Project 2023

San Francisco County Jail # 5

The project provides education services to people who are incarcerated in Bay Area jails.

International Math Olympiad Training Camp 2015, 2017

Homi Bhabha Center for Science Education, Mumbai

I taught problem solving, the probabilistic method, elementary number theory, and basic analytic methods in number theory. I also graded exams.

Undergraduate Student Mentorship.....

Stanford Undergraduate Research in Mathematics 2023

Department of Mathematics, Stanford University

The program is aimed at introducing Stanford undergraduate students to research in mathematics. I mentored Ethan Zhang, Ezra Steinberg, Peter Westbrook, and Quinn McIntyre. We proved several new results related to prophet inequalities. A report is available.

Directed Reading Program 2020–2022, 2024

Department of Mathematics, Stanford University

The program is aimed at helping undergraduates, especially members of historically underrepresented groups, participate in the larger mathematical community. I mentored several students: Alessandra Maranca, Sophia Sanchez, Ellen Xu, Donald Poindexter, Logan Bhamidipaty, and Sohan Vichare.

Graduate Student Mentorship.....

Stanford Summer Engineering Academy 2023

Equity and Inclusion Initiative, Stanford Engineering

I interviewed instructors to hire for the program, helped design a new curriculum, and provided feedback to instructors.

TA Mentoring Program 2022–2023

Department of Mathematics, Stanford University

The program is aimed at providing first-time TAs advice and feedback on leading section. I mentored Yosheb Getachew, Joao Campos Vargas, Zhenyuan Zhang, and Hongjian Yang.

Other Awards

Bronze Medal, 54th International Mathematical Olympiad 2013

KVPY Fellowship 2013–2018

Awarded by the Govt. of India

A 100% scholarship for undergraduate education.

National Talent Search (NTS) Scholarship 2010–2011

Awarded by the Govt. of India to high school students